Giga-Embeddings: Mixture-of-Experts Encoders for High-Throughput Text Embeddings

Anonymous Authors[1]

Abstract

We introduce Giga-Embeddings, a family of text embedding models designed to combine strong retrieval quality with efficient serving. Its largest member is a sparse 10B-parameter Mixture-of-Experts encoder that preserves substantial model capacity while activating only a fraction of it for each token. Across English, Russian, multilingual, and code MTEB benchmarks, the MoE model achieves the strongest aggregate performance within the family on the available evaluations. At an input length of 1024 tokens under our benchmark environment, its throughput is 25% higher than that of the dense 3B model and 1.56–4.37× that of the evaluated external systems. The family further includes a dense 3B model and a distilled 480M model for deployments with tighter compute and memory constraints. Together, these models provide a practical range of quality–efficiency trade-offs for multilingual retrieval and related embedding applications.

1. Introduction

Text embeddings support semantic search, retrieval-augmented generation, clustering, classification, and semantic textual similarity. MTEB (Muennighoff et al., 2023) and its multilingual expansion MMTEB (Enevoldsen et al., 2025) provide broad evaluation suites for these use cases. Recent embedding systems adapt decoder-only language models into bidirectional encoders and train them with large-scale contrastive objectives (BehnamGhader et al., 2024; Wang et al., 2024; Lee et al., 2025a; Kolodin et al., 2025). Although scaling the backbone generally improves representation quality, billion-parameter encoders are costly to store and serve.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Knowledge distillation offers one route to smaller encoders. For embedding models, supervision can be transferred by matching embeddings or by matching the teacher's relative similarity scores over candidate sets (Reimers & Gurevych, 2020; Xu et al., 2023; Zhao et al., 2026). The latter is attractive when the teacher and student have different hidden dimensions or architectures. It is also architecture-agnostic: a dense or sparse Mixture-of-Experts (MoE) teacher can supervise the same listwise objective through its final similarity scores.

Our focus is the quality–throughput trade-off observed for the sparse checkpoint rather than a new MoE-specific distillation algorithm. We characterize the current Giga-Embeddings family, formalize its similarity-distribution objective, and evaluate all available checkpoints under explicit benchmark definitions. The sparse model combines 10B parameters of resident capacity with approximately 1.8B active parameters per token and exceeds the measured throughput of the dense comparison systems. At the compact end, the 480M-parameter checkpoint stays within 2.41–4.06 points of the 3B model on the three suites where both were evaluated. Our contributions are:

1. a sparse 10B-total-parameter encoder with approximately 1.8B active parameters per token and a throughput comparison at three sequence lengths under a shared benchmark environment;
2. a task-macro evaluation of 480M, 3B, and 10B-A1.8B Giga-Embeddings checkpoints on English, Russian, multilingual, and code benchmarks;
3. a reproducible evaluation protocol with task counts, explicit missing coverage, and an official MTEB leaderboard snapshot for external context.

2. Related Work

General-purpose text embeddings. Sentence-BERT (Reimers & Gurevych, 2019) and SimCSE (Gao et al., 2021) established efficient bi-encoder and contrastive recipes. E5 (Wang et al., 2022), GTE (Li et al., 2023), BGE (Xiao et al., 2023), and Nomic Embed (Nussbaum

et al., 2024) scaled data and negative mining. Instruction-conditioned systems such as INSTRUCTOR (Su et al., 2023), E5-Mistral (Wang et al., 2024), GritLM (Muennighoff et al., 2025), and NV-Embed (Lee et al., 2025a) adapt large language models to heterogeneous embedding tasks. Current multilingual LLM-based embedders include Qwen3-Embedding (Zhang et al., 2025), whose backbones the dense checkpoints in our family share, and Gemini Embedding (Lee et al., 2025b); both also rely on distillation or model merging.

Embedding distillation. Classical knowledge distillation matches softened output distributions (Hinton et al., 2015), and compact encoders have long been obtained by direct compression of BERT-style models (Sun et al., 2019; Sanh et al., 2019; Jiao et al., 2020; Sun et al., 2020). In dense retrieval, cross-architecture and listwise similarity distillation are well established: Margin-MSE transfers cross-encoder margins to bi-encoders (Hofstätter et al., 2020), RocketQAv2 applies dynamic listwise distillation between retriever and re-ranker (Ren et al., 2021), and Gecko distills knowledge from large language models into a compact embedder (Lee et al., 2024). For sentence embeddings, Reimers & Gurevych (2020) transfer representations from a monolingual teacher to a multilingual student, while DistillCSE (Xu et al., 2023) studies contrastive self-distillation and teacher-logit variance. GUIDE (Trinh et al., 2025) combines guided initialization with representation transfer. Recent contrastive-plus-distillation recipes for compact embedders include KaLM-Embedding-V2 (Zhao et al., 2026), which weights contrastive loss against a KL divergence between teacher and student similarity distributions, and jina-embeddings-v5-text (Akram et al., 2026), which combines distillation with task-targeted contrastive training. Our objective follows this family and is not claimed as a new loss.

Sparse teachers and dense students. MoE models increase total capacity while activating a subset of experts per token (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024). Sparse-to-dense transfer has been studied for generative models; for example, Xue et al. (2022) compare several ways of gathering expert weights before distillation, and Fedus et al. (2022) describe sparse-to-dense distillation with partial retention of the sparse quality gain. Final-output distillation can use an MoE teacher without exposing its routing decisions, but this alone does not make the objective router- or expert-aware. A controlled study of whether MoE teachers offer benefits over quality-matched dense teachers remains necessary.

3. Model and Training Recipe

3.1. Encoder adaptation

Following the earlier Giga-Embeddings recipe (Kolodin et al., 2025), the model family adapts decoder-only language models for embedding generation. The causal attention mask is replaced by a fully visible bidirectional mask, as in LLM2Vec (BehnamGhader et al., 2024), so that each token can use both left and right context. We use mean pooling to aggregate token states into a fixed-dimensional vector and normalize the resulting vector before cosine-similarity scoring. The instruction templates follow the earlier Giga-Embeddings recipe (Kolodin et al., 2025) and distinguish asymmetric retrieval inputs from symmetric semantic-similarity inputs. The complete instruction set will be released with the model checkpoints.

The exported result metadata identifies embedding dimensions of 1024, 2048, and 1536 for the 480M, 3B, and 10B-A1.8B labels, respectively. Matching checkpoint configurations and weights identify the first two models as dense bidirectional Qwen3 encoders (Zhang et al., 2025) with 480M and 3B parameters. The largest checkpoint is a bidirectional DeepSeekMoE-style encoder (Dai et al., 2024) with 10B total parameters, 64 routed experts, one shared expert, and top-4 routing. Its release designation, A1.8B, denotes approximately 1.8B active parameters per token. We use "480M", "3B", and "10B-A1.8B" as short model-family names.

3.2. Staged contrastive training

The model family uses a staged recipe: broad contrastive pre-training, retrieval fine-tuning with hard negatives, and multitask fine-tuning for retrieval, classification, clustering, and semantic similarity. The training mixture combines open-source datasets used in the earlier Giga-Embeddings recipe (Kolodin et al., 2025) with non-public datasets subject to contractual confidentiality restrictions. The non-public component cannot be enumerated in the paper because of these contractual restrictions. For a batch of B queries, let $\mathcal{C}_i$ be the fixed candidate set for query q_i, let $p(i)$ index its positive candidate, and let s_{ij} be the student cosine similarity. We define the batch-mean contrastive loss, following the InfoNCE objective (van den Oord et al., 2018), as

$$\mathcal{L}_{\mathrm{NCE}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp(s_{i,p(i)}/\tau)}{\sum_{j\in\mathcal{C}_i}\exp(s_{ij}/\tau)}. \qquad (1)$$

Hard negatives are mined from a retrieval rank range; such candidates require qrel filtering or a false-negative audit because rank alone does not guarantee irrelevance.

3.3. Similarity-distribution distillation

The objective below documents the training recipe of the model family as specified in its training code; implementation-level details—the teacher caching scheme, λ, τ_{KD}, and candidate construction—are not recoverable from the evaluation artifacts and are listed in Section 5 as required documentation. Let t_{ij} be the teacher cosine similarity for the same support $\mathcal{C}_i$. Teacher and student distributions are

$$P_T(j \mid q_i) = \frac{\exp(t_{ij}/\tau_{\mathrm{KD}})}{\sum_{k\in\mathcal{C}_i}\exp(t_{ik}/\tau_{\mathrm{KD}})}, \tag{2}$$

$$P_S(j \mid q_i) = \frac{\exp(s_{ij}/\tau_{\mathrm{KD}})}{\sum_{k\in\mathcal{C}_i}\exp(s_{ik}/\tau_{\mathrm{KD}})}. \tag{3}$$

The distillation term is a batch mean over identical supports:

$$\mathcal{L}_{\mathrm{KD}} = \frac{1}{B}\sum_{i=1}^{B} D_{\mathrm{KL}}(P_T(\cdot \mid q_i) \,\|\, P_S(\cdot \mid q_i)). \tag{4}$$

The combined objective is

$$\mathcal{L} = \lambda\mathcal{L}_{\mathrm{KD}} + (1-\lambda)\mathcal{L}_{\mathrm{NCE}}. \tag{5}$$

The combined objective in Equation (5) transfers local listwise similarity structure through Equation (4) while Equation (1) preserves the direct contrastive signal; it does not transfer the teacher's full global embedding geometry. It is also not MoE-aware: router probabilities, expert identities, and expert states do not appear in Equation (4).

Teacher embeddings may be cached and combined into similarities for dynamically formed in-batch supports. Caching only scalar pair scores is valid when $\mathcal{C}_i$ is fixed; it is insufficient for arbitrary shuffled in-batch or cross-device negatives unless all required cross-pairs are materialized. The current evaluation artifacts do not record which caching scheme was used, so this implementation detail remains to be documented from the training code before submission.

4. Evaluation

4.1. Evaluation protocol

Evaluation follows MTEB's canonical task-macro protocol. We use the benchmark metadata distributed with MTEB to select each task's prescribed subsets and evaluation splits, compute one score per task, and average tasks uniformly within a suite. This prevents tasks with more languages, subsets, or splits from receiving additional weight. The accompanying script excludes model metadata, run settings, and notebook checkpoints, and verifies complete task coverage before producing a benchmark aggregate. The evaluated artifacts were frozen on 2026-08-03; SHA-256 digests over every result JSON file are distributed with the aggregation code.

The 3B and 10B-A1.8B task sets match within every common benchmark, so within-column comparisons remain well-defined. A Chinese-language MTEB suite comprising 31 tasks was evaluated only for the 3B checkpoint, where it scores 59.87; it is excluded from the main tables because it lacks cross-checkpoint coverage and is included here for completeness. The Russian suite follows ruMTEB (Snegirev et al., 2025), while the multilingual suite follows MMTEB (Enevoldsen et al., 2025). The aggregation pipeline is identical and model-independent across the evaluated checkpoints.

Theoretical fp16 weight storage is approximately 0.96, 6, and 20 GB for the 480M, 3B, and 10B-A1.8B checkpoints, respectively (decimal GB, excluding runtime state). Thus the distilled checkpoint's fp16 weights occupy about $1/20.8$ of the MoE reference's resident-weight storage (0.96 vs. 20 GB). This is a parameter-derived storage estimate, not a measurement of peak serving memory, latency, or throughput.

4.2. Aggregate results

Table 2 presents task-macro scores for all three checkpoints. The 480M distilled model reaches 69.52 on English, 70.96 on Russian, and 72.87 on Code. Relative to the 3B checkpoint, its gaps are 2.41, 3.60, and 4.06 points, respectively. Multilingual was not evaluated for the 480M model—the run was never executed, not withheld—so we do not estimate or impute that cell. We caution that the 480M-versus-3B gap on Multilingual could be materially larger than the 2.41–4.06-point range observed on the evaluated suites; the compression comparison is therefore bounded to English, Russian, and Code.

The 10B-A1.8B checkpoint is numerically strongest on every benchmark for which it was evaluated, though the English and Russian margins fall in the sub-one-point range for which we lack uncertainty estimates (Section 5). The improvement over 3B is modest on English (+0.30) and Russian (+0.42), and larger on Multilingual (+1.75) and Code (+1.48). This pattern is consistent with diminishing aggregate returns on the two language-specific suites, but it is not a controlled scaling law: the 3B and 10B-A1.8B checkpoints were produced by different training recipes, the MoE activates fewer parameters per token ($\sim$1.8B) than the 3B dense model, and their embedding dimensions are not monotonic in nominal model scale.

The scores also do not isolate the benefit of distillation. Results from a 480M checkpoint trained for the same number of updates on the same data without teacher supervision are not yet available. Likewise, the 3B and 10B-A1.8B results are not substitutes for a size-matched non-distilled control. We therefore treat the 480M score as the quality of an evaluated distilled checkpoint, not as an estimate of the treatment

Table 1. Architecture summary. Parameter counts use rounded release-scale designations; "Active" is per token.

Type	Total	Active	Dim.
Dense	480M	480M	1024
Dense	3B	3B	2048
MoE, top-4	10B	∼1.8B	1536

Table 2. Task-macro MTEB scores (%) over 41 English, 23 Russian, 131 multilingual, and 12 code tasks. Bold marks the best result within the model family; a dash denotes a missing evaluation.

Scale	Dim.	English	Russian	Multilingual	Code
480M	1024	69.52	70.96	—	72.87
3B	2048	71.93	74.56	63.89	76.93
10B-A1.8B	1536	**72.23**	**74.98**	**65.64**	**78.41**

Table 3. Token throughput under one shared benchmark environment. Values are thousands of tokens per second at fixed input lengths; the backend is shown with each model. The final column is each model's throughput at 1024 tokens divided by the throughput of our 10B-A1.8B-0826 checkpoint. Because backends differ, the table compares complete serving stacks rather than model architectures in isolation.

Model / backend	512 tokens	1024 tokens	2048 tokens	Relative
Qwen3 Embedding 4B / vLLM	70.1k	73.2k	71.2k	0.64×
NV-Embed-v2 / Transformers	25.6k	26.2k	25.6k	0.23×
F2LLM-v2-8B / vLLM	43.2k	43.4k	42.6k	0.38×
Nemotron 8B / vLLM	42.6k	43.2k	41.7k	0.38×
Ours: 3B-0826 / vLLM	87.9k	91.5k	90.4k	0.80×
Ours: 10B-A1.8B-0826 / vLLM	**112.6k**	**114.5k**	**102.3k**	**1.00**×

effect of distillation.

4.3. Inference throughput

The MoE serving stack has the highest measured throughput at every evaluated length. Relative to the dense Giga-Embeddings 3B checkpoint, its throughput is 1.28×, 1.25×, and 1.13× as high at 512, 1024, and 2048 tokens, respectively. These measurements compare complete checkpoints and serving stacks and do not isolate the causal contribution of sparse activation. Throughput was not measured for the 480M checkpoint; its deployment comparison in this paper is therefore limited to theoretical weight storage.

4.4. External baseline comparison

The external rows in Table 4 come from the official MTEB Leaderboard rather than new local inference runs. They provide a reproducible snapshot of public results, but differences in model-specific prompts and evaluation settings make them contextual comparisons rather than a controlled same-run experiment. The raw API responses and a dependency-free parser are included with the supplementary material. Table 5 provides the corresponding comparison for the two larger Giga-Embeddings checkpoints. Giga-Embeddings-instruct-10B-A1.8B-0826 leads the Russian column, while Qwen3-Embedding-8B leads English and Multilingual and F2LLM-v2-14B leads Code.

5. Evidence Boundaries and Required Controls

Distillation and teacher architecture. A causal distillation study requires a 480M no-KD control with identical data, candidate sets, optimizer, and update count. To study MoE teachers specifically, one additionally needs quality- or compute-matched dense and MoE teachers and the same student initialization. Router- or expert-aware variants should only be described as such if routing signals enter the objective. The current evaluation artifacts and checkpoint names do not contain a run manifest that cryptographically links the evaluated 10B-A1.8B checkpoint to the 480M training run; that teacher–student provenance must be added before the MoE model is called the exact teacher in an archival paper.

Statistical reliability. Each checkpoint has a single evaluation run. Future experiments should include multiple training seeds or paired task-level bootstrap intervals, especially for aggregate differences below one point. The full per-task scores will be released with the model artifacts. The outstanding 480M measurements are the matched no-KD control and the Multilingual evaluation, which are not included in the current comparison.

Training-data overlap. Multitask fine-tuning can overlap with MTEB train, development, or test collections. A final submission must list the public datasets and characterize restricted sources at the level permitted by their contractual terms, disclose overlap, deduplicate training examples against evaluation text, and provide a no-overlap aggregate. Teacher pre-training and synthetic-data provenance should

Table 4. External comparison on MTEB benchmark suites (%). External rows are official MTEB Leaderboard `meanTask` values retrieved on 2026-08-08; our 480M-0826 row uses the evaluation from Table 2. Bold scores mark the best result in each column. A dash means that no complete benchmark score is available in the corresponding source.

Model	Params	Dim.	Russian	English	Multilingual	Code
sergeyzh/BERTA	128M	768	69.39	—	—	—
google/embeddinggemma-300m	308M	768	65.19	69.67	61.15	68.76
codefuse-ai/F2LLM-v2-0.6B	596M	1024	65.97	69.97	62.74	**77.41**
Qwen/Qwen3-Embedding-0.6B	596M	1024	64.29	**70.47**	**64.34**	75.42
ai-forever/FRIDA	823M	1536	70.95	—	—	—
Ours: 480M-0826	480M	1024	**70.96**	69.52	—	72.87

Table 5. Comparison of our 3B and 10B-A1.8B checkpoints with larger embedding baselines (%). External rows are official MTEB Leaderboard `meanTask` values retrieved on 2026-08-08; our two rows use the evaluation from Table 2. Bold scores mark the best result in each column. A dash means that no complete benchmark score is available in the corresponding source.

Model	Params	Dim.	Russian	English	Multilingual	Code
ai-forever/FRIDA	823M	1536	70.95	—	—	—
codefuse-ai/F2LLM-v2-1.7B	1.72B	2048	68.52	71.63	65.21	78.76
codefuse-ai/F2LLM-v2-4B	4.02B	2560	69.46	72.41	67.06	80.15
Qwen/Qwen3-Embedding-4B	4.02B	2560	70.13	74.61	69.45	80.07
Qwen/Qwen3-Embedding-8B	7.57B	4096	71.42	**75.23**	**70.58**	80.69
codefuse-ai/F2LLM-v2-8B	7.57B	4096	70.57	72.86	68.09	80.16
codefuse-ai/F2LLM-v2-14B	13.99B	5120	70.90	73.08	68.74	**80.75**
Ours: 3B-0826	3B	2048	74.56	71.93	63.89	76.93
Ours: 10B-A1.8B-0826	10B	1536	**74.98**	72.23	65.64	78.41

be included in the same audit.

Efficiency. Table 3 measures token throughput under one shared benchmark environment, but it does not characterize latency, energy, or peak memory and should not be extrapolated to other hardware or workloads. The complete benchmark release must specify hardware and GPU count, precision, batch size, concurrency, warmup, backend versions, and the timing harness. Resident and active parameters, FLOPs, peak memory, latency, and energy provide complementary views of efficiency.

Reproducibility and governance. Exact checkpoint identifiers, remaining architecture details, releasable dataset metadata, filtering, and optimizer schedules are not recoverable from the evaluation artifacts alone. They remain required for an archival version, together with a discussion of contractual data restrictions, data licenses, PII handling, and language coverage.

6. Conclusion

Giga-Embeddings-10B-A1.8B combines 10B parameters of resident MoE capacity with approximately 1.8B active parameters per token. It is the fastest of six systems in our serving comparison and gives the best observed aggregate quality within the Giga-Embeddings family. The 480M distilled checkpoint remains within 2.41–4.06 points of the 3B model on English, Russian, and Code, while using about $1/20.8$ of the MoE checkpoint's theoretical fp16 weight storage; its Multilingual score remains unmeasured. Quality gains from 3B to 10B-A1.8B are small on English and Russian but more visible on Multilingual and Code. Under the evaluated serving setup, the MoE checkpoint combines the highest measured throughput with the strongest aggregate scores within the model family, while causal claims about sparse activation, distillation, and MoE teachers still require matched controls.

References

Akram, M. K., Sturua, S., Havriushenko, N., Herreros, Q., Günther, M., Werk, M., and Xiao, H. jina-embeddings-v5-text: Task-targeted embedding distillation. *arXiv preprint arXiv:2602.15547*, 2026.

BehnamGhader, P., Adlakha, V., Mosbach, M., Bahdanau, D., Chapados, N., and Reddy, S. LLM2Vec: Large language models are secretly powerful text encoders. In *Proceedings of the First Conference on Language Modeling (COLM)*, 2024.

Dai, D., Deng, C., Zhao, C., Xu, R. X., Gao, H., Chen, D., Li, J., Zeng, W., Yu, X., Wu, Y., et al. DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models. In *Proceedings of the 62nd Annual*

Meeting of the Association for Computational Linguistics (ACL), pp. 1280–1297, 2024.

Enevoldsen, K., Chung, I., Kerboua, I., Kardos, M., Mathur, A., Stap, D., Gala, J., Siblini, W., Krzemiński, D., Winata, G. I., et al. MMTEB: Massive multilingual text embedding benchmark. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2502.13595.

Fedus, W., Zoph, B., and Shazeer, N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

Gao, T., Yao, X., and Chen, D. SimCSE: Simple contrastive learning of sentence embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6894–6910, 2021.

Hinton, G., Vinyals, O., and Dean, J. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Hofstätter, S., Althammer, S., Schröder, M., Sertkan, M., and Hanbury, A. Improving efficient neural ranking models with cross-architecture knowledge distillation. *arXiv preprint arXiv:2010.02666*, 2020.

Jiang, A. Q., Sablayrolles, A., Roux, A., Mensch, A., Savary, B., Bamford, C., Chaplot, D. S., de las Casas, D., Hanna, E. B., Bressand, F., Lengyel, G., Bour, G., Lample, G., Lavaud, L. R., Saulnier, L., Lachaux, M.-A., Stock, P., Subramanian, S., Yang, S., Antoniak, S., Scao, T. L., Gervet, T., Lavril, T., Wang, T., Lacroix, T., and Sayed, W. E. Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024.

Jiao, X., Yin, Y., Shang, L., Jiang, X., Chen, X., Li, L., Wang, F., and Liu, Q. TinyBERT: Distilling BERT for natural language understanding. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pp. 4163–4174, 2020.

Kolodin, E., Khomich, D., Savushkin, N., Ianina, A., and Minkin, F. GigaEmbeddings: Efficient Russian language embedding model. In *Proceedings of the 10th Workshop on Slavic Natural Language Processing (SlavicNLP 2025)*, pp. 17–24, 2025.

Lee, C., Roy, R., Xu, M., Raiman, J., Shoeybi, M., Catanzaro, B., and Ping, W. NV-Embed: Improved techniques for training LLMs as generalist embedding models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025a.

Lee, J., Dai, Z., Ren, X., Chen, B., Cer, D., Cole, J. R., Hui, K., Boratko, M., Kapadia, R., Ding, W., et al. Gecko: Versatile text embeddings distilled from large language models. *arXiv preprint arXiv:2403.20327*, 2024.

Lee, J., Chen, F., Dua, S., Cer, D., Shanbhogue, M., Naim, I., Ábrego, G. H., Li, Z., Chen, K., Vera, H. S., et al. Gemini embedding: Generalizable embeddings from Gemini. *arXiv preprint arXiv:2503.07891*, 2025b.

Li, Z., Zhang, X., Zhang, Y., Long, D., Xie, P., and Zhang, M. Towards general text embeddings with multi-stage contrastive learning. *arXiv preprint arXiv:2308.03281*, 2023.

Muennighoff, N., Tazi, N., Magne, L., and Reimers, N. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pp. 2014–2037, 2023.

Muennighoff, N., Su, H., Wang, L., Yang, N., Wei, F., Yu, T., Singh, A., and Kiela, D. Generative representational instruction tuning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.

Nussbaum, Z., Morris, J. X., Duderstadt, B., and Mulyar, A. Nomic embed: Training a reproducible long context text embedder. *arXiv preprint arXiv:2402.01613*, 2024.

Reimers, N. and Gurevych, I. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 3982–3992, 2019.

Reimers, N. and Gurevych, I. Making monolingual sentence embeddings multilingual using knowledge distillation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 4512–4525, 2020. doi: 10.18653/v1/2020.emnlp-main.365.

Ren, R., Qu, Y., Liu, J., Zhao, W. X., She, Q., Wu, H., Wang, H., and Wen, J.-R. RocketQAv2: A joint training method for dense passage retrieval and passage re-ranking. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 2825–2835, 2021.

Sanh, V., Debut, L., Chaumond, J., and Wolf, T. DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q. V., Hinton, G. E., and Dean, J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017.

Snegirev, A., Tikhonova, M., Maksimova, A., Fenogenova, A., and Abramov, A. The Russian-focused embedders' exploration: ruMTEB benchmark and Russian embedding model design. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 236–254, 2025. doi: 10.18653/v1/2025.naacl-long.12.

Su, H., Shi, W., Kasai, J., Wang, Y., Hu, Y., Ostendorf, M., tau Yih, W., Smith, N. A., Zettlemoyer, L., and Yu, T. One embedder, any task: Instruction-finetuned text embeddings. In *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 1102–1121, 2023.

Sun, S., Cheng, Y., Gan, Z., and Liu, J. Patient knowledge distillation for BERT model compression. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 4323–4332, 2019.

Sun, Z., Yu, H., Song, X., Liu, R., Yang, Y., and Zhou, D. MobileBERT: a compact task-agnostic BERT for resource-limited devices. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 2158–2170, 2020.

Trinh, K., Menghani, G., and Vee, E. GUIDE: Guided initialization and distillation of embeddings. *arXiv preprint arXiv:2510.06502*, 2025.

van den Oord, A., Li, Y., and Vinyals, O. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., and Wei, F. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*, 2022.

Wang, L., Yang, N., Huang, X., Yang, L., Majumder, R., and Wei, F. Improving text embeddings with large language models. *arXiv preprint arXiv:2401.00368*, 2024.

Xiao, S., Liu, Z., Zhang, P., Muennighoff, N., Lian, D., and Nie, J.-Y. C-Pack: Packed resources for general Chinese embeddings. *arXiv preprint arXiv:2309.07597*, 2023.

Xu, J., Shao, W., Chen, L., and Liu, L. DistillCSE: Distilled contrastive learning for sentence embeddings. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 8153–8165, 2023. doi: 10.18653/v1/2023.findings-emnlp.547.

Xue, F., He, X., Ren, X., Lou, Y., and You, Y. One student knows all experts know: From sparse to dense. *arXiv preprint arXiv:2201.10890*, 2022.

Zhang, Y., Li, M., Long, D., Zhang, X., Lin, H., Yang, B., Xie, P., Yang, A., Liu, D., Lin, J., et al. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*, 2025.

Zhao, X., Hu, X., Shan, Z., Huang, S., Zhou, Y., Zhang, X., Sun, Z., Liu, Z., Li, D., Wei, X., Pan, Y., Xiang, Y., Zhang, M., Wang, H., Yu, J., Hu, B., and Zhang, M. KaLM-Embedding-V2: Superior training techniques and data inspire a versatile embedding model. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.